Tab 1

## 2.3 ANOVA

*[Provide the full name of the team member responsible for preparing the study to collect data that will be studied with Independent-sample-t-test.*
*]*

***Student Name: Saad***

| | | |
|---|---|---|
| **Research question** | *Do users with different levels of flight-booking experience (**novice**, **intermediate**, and **expert**) differ significantly in the **number of filters** they apply when searching for flights on Google Flights?* | |
| **Usability metric studied** | ***Efficiency and Task Strategy – Measured by Number of Filters Used***<br>*The metric captures how effectively users employ filtering tools (e.g., "Stops," "Price," "Times," "Airlines") to locate a desired flight within constraints such as price or duration.* | |
| **Hypothesis (statement)** | ***Null Hypothesis ($H_0$):** There is no significant difference in the average number of filters used between novice, intermediate, and expert users when searching for flights on Google Flights.* | |
| | ***Alternative Hypothesis ($H_1$):** There is a significant difference in the average number of filters used between at least two of the experience groups (novice, intermediate, expert).* | |
| **Mathematical Expression** | ***Null Hypothesis ($H_0$):** $\mu_1 = \mu_2 = \mu_3$* | |
| | ***Alternative Hypothesis ($H_1$):** At least one group mean differs, i.e. $\mu_1 \neq \mu_2$ or $\mu_1 \neq \mu_3$ or $\mu_2 \neq \mu_3$*<br>*where:*<br>*$\mu_1$ = mean number of filters used by novice participants*<br>*$\mu_2$ = mean number of filters used by intermediate participants*<br>*$\mu_3$ = mean number of filters used by expert participants* | |
| **Variables** | *Independent* | *User experience level*<br>*Three categories – Novice (0–2 bookings), Intermediate (3–10 bookings), Expert (>10 bookings)* |
| | *Dependent* | *Number of filters used*<br>*Count of filters applied while completing the flight search task* |
| | *Control* | *Flight route and dates (London → Rome, 10–17 Dec), device and browser, environment, and task instructions* |

| | | |
|---|---|---|
| | *Measurement Scale* | ***Number of filters used:*** *Ratio scale (numeric count).*<br><br>***Optional supporting metric:*** *Task completion time (seconds, ratio scale) for context.* |
| | *Data type* | ***Quantitative*** *— numerical count data representing the number of filters applied by each participant.* |
| **Experimental design** | ***Between-subjects (one-way ANOVA)***<br><br>*Each participant belongs to only one experience group (novice, intermediate, or expert). A between-subjects design was selected to compare mean filter use across independent groups. This approach minimises carry-over effects and ensures that differences are attributable to user experience rather than practice or fatigue.* | |
| **Tasks and Scenarios** | *All participants perform the same task described above (London → Rome search). The scenario simulates a realistic goal-oriented booking process where participants need to find a suitable flight within budget constraints. This allows observation of filter use behaviour under identical conditions, making group comparisons valid for ANOVA analysis.* | |
| **Procedure** | *The experiment was conducted* ***in person*** *in a quiet computer lab and was* ***moderated*** *to ensure consistency and accurate timing.*<br><br>1. ***Introduction (5 min):*** *Participants were welcomed, briefed on the study purpose, and asked to sign a consent form (see Appendix I – Participant Information Sheet).*<br><br>2. ***Pre-task survey (3 min):*** *Participants reported their past online flight bookings to determine experience level.*<br><br>3. ***Task (10–12 min):*** *Participants performed the flight search task while thinking aloud. The moderator recorded start and end times and observed filter use.*<br><br>4. ***Post-task questionnaire (5 min):*** *Participants rated how easy they found the filter feature on a 5-point Likert scale.*<br><br>5. ***Debrief (5 min):*** *Participants were thanked and informed of data confidentiality.*<br><br>*Each session lasted approximately* ***25–30 minutes****, and data collection for all 20+ participants was completed within one week.* | |

| | |
|---|---|
| **Data collection method** | ***Tools Used:***<br><br>● ***OBS Studio*** *for screen recording and timestamp verification.*<br><br>● ***Digital timer*** *to record task duration in seconds.*<br><br>● ***Observation sheet*** *to log the number of filters applied and notable behaviour.*<br><br>● ***Google Form*** *for pre-task and post-task questionnaires.*<br><br>***Justification:*** *OBS ensures accurate capture of filter use and timing, while Google Forms automatically export data for statistical analysis in Excel or SPSS.* |
| **Test material** | ***Materials:***<br><br>1. *Printed/digital task instructions and participant brief.*<br><br>2. *Pre-task survey to classify experience level.*<br><br>3. *Observation sheet for manual data logging.*<br><br>4. *Post-task questionnaire for perceived ease rating.*<br><br>***Measurement Definition:***<br><br>● ***Task start:*** *When participant clicks "Search" on Google Flights.*<br><br>● ***Task end:*** *When participant verbally indicates they have found a suitable flight matching criteria.*<br><br>● *Moderator records the total time and number of filters used.* |

Absolutely — here are the two things you asked for:

---

# ✅ Appendix Materials (Surveys, Checklists, Task Sheets)

These are formatted exactly the way coursework appendices are expected to look.

---

# 📎 Here is a detailed, polished, academic-style Task Instruction Sheet for Appendix II – ANOVA Study (Saad).
# Clear, structured, and ready to paste directly into your report.

---

# Appendix II — Task Instruction Sheet (Saad – ANOVA Study)

**Task Title:**

# Flight Search Task – Google Flights (Error Rate Measurement for ANOVA Study)

---

**Scenario**

**You are planning a round-trip flight from Sri Lanka to London for a one-week stay. Your goal is to find a flight that meets all the following criteria:**

- **Departure: Sri Lanka → London**
- **Travel Dates: 10 December (departure) to 17 December (return)**
- **Maximum Budget: £250**
- **Maximum Stops: 1 stop only (for both outbound and return flights)**
- **Platform: Google Flights (desktop version only)**

**This scenario replicates a realistic travel planning situation where cost, dates, and**

**stopovers must be balanced to select a suitable itinerary.**

---

### Your Task

**You must use Google Flights to locate a valid round-trip itinerary that matches every requirement listed above.**
**As you perform the task, you must share your screen continuously so the moderator can observe your actions and record error occurrences accurately.**

---

### Task Requirements (All must be satisfied)

**Please ensure your final selected itinerary strictly matches *every* criterion:**

**1. Route**

**✓ Sri Lanka → London (round trip)**

**2. Dates**

✓ Outbound: 10 December
✓ Return: 17 December

**3. Price**

✓ Total fare must be £250 or below

**4. Stops**

✓ Maximum 1 stop on each leg
❌ No itineraries with 2+ stops

**5. Platform**

✓ Google Flights (desktop/laptop browser only)
❌ Do not use mobile devices or third-party websites

**6. Task Behaviour**

✓ Share your screen throughout
✓ Do not ask for help or hints
✓ Perform the task as naturally as possible

---

**Task Start and Timing**

**Task Start:**
**The task officially begins at the moment you click “Search” after entering your route and travel dates.**

**What you should say:**
**➞ “I’m starting now.”**

**The moderator will start the timer at that point.**

---

### Task End

**The task ends only when you verbally confirm that you have selected what you believe to be a valid itinerary.**

**What you should say:**
**➞ “I have found the itinerary.”**

**The moderator will then:**

- **Stop the timer**

- **Verify whether the selected itinerary satisfies all criteria**

- **Review the screen recording afterward for error checking**

---

**Important Notes for Participants**

- **You may navigate the site freely; no assistance will be provided.**

- **Errors such as choosing the wrong dates, exceeding budget, or selecting flights with too many stops will be recorded using the Error Checklist.**

- **Your performance is not graded; only the usability of Google Flights is being evaluated.**

- **You may ask clarification questions *before* the task starts, but not during**

**the task.**

---

### End of Task Instructions

**Thank you for participating in this usability study. Your interaction will contribute to valuable insights for our ANOVA analysis on user experience levels and error behaviour.**

---

**If you want, I can also prepare:**

- 📌 **Appendix III – Pre-Test Survey**
- 📌 **Appendix IV – Error Checklist**
- 📌 **Appendix V – Post-Test Questionnaire**

**Just say: “Give me Appendix III/IV/V.”**

---

## 🖇️ Appendix III — Pre-Task Survey (Experience Classification)

**Google Flights – Pre-Test Questionnaire**

1. **What is your age group?**
   ▢ 18–25
   ▢ 26–35
   ▢ 36–50
   ▢ 51+

2. **How many times have you booked flights online before?**
   ▢ 0–2 times (Novice)
   ▢ 3–10 times (Intermediate)
   ▢ More than 10 times (Expert)

3. **Which platforms have you previously used for booking flights?**
   (Select all that apply)
   ▢ Google Flights
   ▢ Skyscanner
   ▢ Expedia
   ▢ Airline Websites
   ▢ Other: _____

4. **How familiar are you with Google Flights?**
   1 = Not at all familiar → 5 = Very familiar
   ▢ 1 ▢ 2 ▢ 3 ▢ 4 ▢ 5

5. **Do you consent to screen-sharing and participation in this moderated study?**
   ▢ Yes
   ▢ No

---

# Appendix IV — Moderator Error Checklist (Primary DV)

## Error Checklist – Google Flights ANOVA Study

Moderator: _____
Participant ID: _____
Experience Group: □ Novice □ Intermediate □ Expert

| Error Type | Description | Occurred? (✓) | Notes |
|---|---|---|---|

| | | | |
|---|---|---|---|
| **E1: Wrong Departure Date** | Did not enter 10 Dec | ☐ | |
| **E2: Wrong Return Date** | Did not enter 17 Dec | ☐ | |
| **E3: Wrong Destination** | Selected incorrect airport/city | ☐ | |
| **E4: Violated Budget (£250)** | Chose flight > £250 | ☐ | |
| **E5: Wrong Number of Stops** | Selected >1 stop | ☐ | |
| **E6: Failed to Apply Filters Correctly** | Missed price/stop filters | ☐ | |
| **E7: Chose Wrong Flight Option** | Selected unrelated itinerary | ☐ | |
| **E8: Mis-clicks / Navigation Errors** | E.g., backtracking unnecessarily | ☐ | |
| **E9: Incomplete Task** | Could not find valid itinerary | ☐ | |
| **E10: Other Errors** | Any additional mistakes | ☐ | Notes: _____ |

**Total Error Count: _____**

---

# 📎 Appendix V — Post-Test Questionnaire (Error Study)

## Google Flights – Post-Task Survey (Saad – ANOVA)

1. **How difficult was it to complete the task?**
   1 = Very Easy → 5 = Very Difficult
   ▢ 1 ▢ 2 ▢ 3 ▢ 4 ▢ 5

2. **How confident were you in your final selection?**
   1 = Not confident → 5 = Very confident
   ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

3. **Did you feel the task instructions were clear?**
   ☐ Yes
   ☐ No

4. **Did you encounter any technical issues?**
   ☐ Yes
   ☐ No
   If yes, explain: _____

5. **Any comments about your experience?**

---

# ✅ Chapter 4 – ANOVA Interpretation Section

Below is a **full, polished, academically sound interpretation** (ready for Chapter 4), assuming you will insert your actual SPSS/Excel results later.

You can paste your actual F-value, p-value, and means into the placeholders.

---

# 4.3 ANOVA Results and Interpretation (Saad Mazhar)

The purpose of this ANOVA test was to examine whether user experience level (Novice, Intermediate, Expert) had a significant effect on **Error Count** while completing the Google Flights task.

### 4.3.1 Descriptive Statistics (Group Means)

(Insert your actual values here)

| Group | Mean Error Count | SD |
|---|---|---|
| Novice | M = ___ | SD = ___ |

| | | |
|---|---|---|
| Intermediate | M = ___ | SD = ___ |
| Expert | M = ___ | SD = ___ |

Preliminary observation suggests that novices produced more errors than intermediate and expert users.

---

# 4.3.2 ANOVA Output

A one-way ANOVA was conducted to compare the effect of flight-booking experience level on total error count.

**ANOVA Summary:**

- **F(2, N-3) = ____**
- **p = ____**
- **$\eta^2$ (effect size) = ____**

(Replace values once you compute them.)

---

# 4.3.3 Interpretation of Results

### If p < 0.05 (Significant):

The ANOVA revealed a **significant effect** of flight-booking experience on error count, indicating that at least one experience group differed from the others.

Post-hoc tests (Tukey HSD) showed:

- Novices made significantly more errors than Intermediate users
- Novices made significantly more errors than Experts
- Intermediate and Expert users did not significantly differ

This suggests that **greater real-world flight-booking experience reduces the likelihood of errors when interacting with Google Flights**, indicating strong learnability and familiarity effects.

---

### If p > 0.05 (Not Significant):

The ANOVA showed **no significant difference** in error count between novice, intermediate, and expert groups.

This indicates that **experience level does not strongly influence error likelihood**, suggesting that Google Flights' interface may offer equal usability across diverse user backgrounds, or that the task was simple enough for all groups to perform similarly.

---

# 4.3.4 Conclusion

Overall, the ANOVA analysis provides insight into how experience level affects (or does not affect) user errors when performing flight-search tasks. These findings help evaluate whether Google Flights supports users with varying levels of prior knowledge and contributes to understanding its error tolerance usability dimension.

---

If you'd like, I can also provide:

📌 A **perfectly formatted Chapter 4 ready to submit**
📌 SPSS/Excel ANOVA table templates
📌 A graph (bar chart with error bars) for your results

Just tell me!

Tab 2

### 3.1.3 Experimental Design for Research Question 3 (Saad)

The experimental design chosen for this study is a **within-subjects design**, also known as a **repeated-measures design**. This design was selected to evaluate the usability impact of using the **Flexible Dates calendar** compared with the **Standard Date search** on Google Flights, specifically in terms of **task completion time** when finding the cheapest round-trip flight.

In a within-subjects design, all participants perform both conditions — in this case, completing the same flight search task twice: once using the standard date search and once using the flexible dates feature. This approach allows each participant to serve as their own control, thereby eliminating individual differences such as search experience, familiarity with Google products, or general decision-making speed.

The dependent variable measured is **task completion time (in seconds)**, while the independent variable is the **type of search method** used (Standard Date vs Flexible Dates). To reduce **order effects** (e.g., learning or fatigue), the order of task presentation was **counterbalanced** across participants — half of the participants started with the standard date task, and the other half started with the flexible dates task.

This design was selected because it offers greater **statistical power** with a smaller sample size and directly compares the two search methods under consistent participant conditions. The data collected from both conditions for each participant will be analysed using a **paired-sample t-test**, which is appropriate for comparing the mean performance across two related measurements.

In summary, the within-subjects design ensures a controlled, reliable comparison between the two Google Flights search methods, minimising between-participant variability while efficiently addressing the research question on task efficiency and usability.

3.2.3

### 3.2.3 ANOVA – In-Person (Moderated) Task (Saad)

To answer the research question, *"Do users with different levels of flight-booking experience (novice, intermediate, and expert) differ significantly in the number of filters they use on Google Flights?"*, an **in-person unmoderated task** design was implemented.

**Procedure:**

Participants were grouped according to their self-reported experience level with booking flights online. Each participant received **clear written instructions** outlining the objective and steps of the task. They were asked to use **Google Flights** to find a **round-trip flight from London to Rome** between **10 and 17 December**, within a **£250 budget** and limited to **one stopover**.

The tasks were conducted **physically in a lab setting** but were **unmoderated** to minimise researcher influence and allow participants to work independently. This approach helped capture authentic user behaviour and natural interaction with Google Flights' filtering features.

After completing the task, participants filled out a **post-task survey** indicating the **number of filters used** (e.g., Stops, Price, Airlines, Times) and rating their ease of use on a **5-point Likert scale**. The self-reported data was validated through **screen recordings** that logged filter interactions and completion times.

Each session lasted approximately **25–30 minutes**, including time for reading instructions, completing the task, and submitting post-task feedback. Data was collected from **over 20 participants** across all experience levels during a one-week testing period.

**Justification:**

An **unmoderated in-person setup** was chosen to ensure participants interacted with the Google Flights interface naturally, without pressure or bias from a facilitator. Standardised written instructions and identical devices were used to maintain consistency and ensure data accuracy.

**Data Collection Tools and Materials:**

1. **Screen recording software (OBS Studio)** – to verify filter usage and completion time.

2. **Post-task survey** – to collect filter counts and perceived ease ratings.

3. **Written task instructions** – provided in both printed and digital formats (see *Appendix II. Task Instruction Sheet*).

---

The task designed for Saad's hypothesis employed a **one-way between-subjects ANOVA** to investigate whether **participants' level of experience with flight booking systems** (novice, intermediate, or expert) significantly influenced their **use of filters** when searching for flights on Google Flights.

**Procedure**

All testing sessions were conducted **in person** and **moderated** to maintain consistency across participants and ensure accurate timing and observation. Each participant was welcomed and briefed about the study's purpose, confidentiality, and procedure, followed by signing the consent form (see Appendix I – Participant Information Sheet). The moderator followed a standardized script to provide identical instructions to each participant.

Participants were first classified into one of three **experience categories** based on a short pre-task questionnaire:

- **Novice:** 0–2 prior flight bookings online
- **Intermediate:** 3–10 prior flight bookings online
- **Expert:** more than 10 prior flight bookings online

After classification, participants completed a single standardized task on **Google Flights** using the same route, dates, and conditions to ensure comparability.

**Task Description**

**Task:**

> "You are planning a round-trip from London to Rome between 10 and 17 December. Your budget is £250, and you prefer a flight with at most one stopover. Using Google Flights, apply the necessary filters and find a suitable flight that meets these conditions."

The moderator instructed participants to think aloud as they completed the task. Once the participant indicated they had found an acceptable flight, the moderator stopped the timer and recorded the number of filters used, time taken, and any observable behaviour patterns.

**Data Collected**

- **Dependent Variable:** Number of filters applied (count)
- **Independent Variable:** Experience level (Novice / Intermediate / Expert)

- **Control Variables:**
    - Same origin/destination and dates
    - Same browser and device
    - Identical instructions and environment

The **number of filters** used (e.g., price, stops, airlines, times) was recorded manually by the moderator using an observation checklist. Additional qualitative notes were made on participant strategy and behaviour to support the quantitative findings.

**Test Duration and Timing**

Each session lasted approximately **25–30 minutes** in total, including:

- 5 minutes for participant welcome, consent, and briefing
- 10–12 minutes for the search task and observation
- 5 minutes for post-task questionnaire and short interview
- 5–8 minutes buffer between participants

The overall data collection period spanned **one week**, during which a minimum of **20 participants** were tested across the three experience levels.

**Standardisation and Consistency**

To ensure consistent data collection:

- All participants received identical verbal instructions and used the same version of the Google Flights interface.
- The moderator used a pre-defined observation sheet to record time, number of filters, and task completion confirmation.
- Environmental conditions (lighting, internet speed, and device type) were kept constant throughout testing sessions.

This structured procedure ensured that any observed differences in filter use could be confidently attributed to participants' **experience level**, supporting valid ANOVA analysis for the hypothesis.

# How to 3.2.3

# 🧠 Step 1. Clarify Your Research Setup

## Goal

Find out if **user experience level** (novice / intermediate / expert) changes how many **filters** people use when searching for flights on **Google Flights**.

## Variables

- **Independent variable (IV):** Experience level (3 groups: novice, intermediate, expert)
- **Dependent variable (DV):** Number of filters used during a search (numeric count)
- **Control variables:**
    - Same task (London → Rome, 10–17 Dec)
    - Same device/browser/internet
    - Same room and moderator
    - Same instructions

This setup gives you 3 groups → perfect for a **one-way between-subjects ANOVA**.

---

# 👥 Step 2. Recruit Participants

You need at least **20 participants total** — ideally divided roughly evenly across the 3 experience levels.

| Experience | Definition | Target count |
|---|---|---|
| Novice | 0–2 flight bookings online | ~7 |
| Intermediate | 3–10 flight bookings | ~7 |

| | | |
|---|---|---|
| Expert | 10+ flight bookings | ~6 |

### How to find them

- Ask classmates, friends, or family who have used flight sites different numbers of times.
- In the recruitment message, mention it's a **15–20 minute usability study on Google Flights**.

Save evidence (emails or messages) — this goes in your appendix.

---

## 📜 Step 3. Prepare Documents

You'll need three key documents **before testing**:

1. **Participant Information Sheet** — explains purpose, duration, voluntary participation, data handling, confidentiality.
2. **Consent Form** — includes checkboxes for consent to record and anonymise data.
3. **Pre-task Questionnaire** — asks how often they've booked flights (to classify them into novice/intermediate/expert).

👉 Bring printed copies, or digital versions if testing remotely.

---

## 💻 Step 4. Set Up Your Environment

- **Device:** Laptop or desktop (use the same one for all sessions).
- **Browser:** Google Chrome (same version).
- **Website:** https://www.google.com/flights
- **Recording tool:** Use **OBS Studio** (free screen recorder).

- **Observation sheet:** Prepare a simple table:

| Participant ID | Experience | Time started | Time ended | Time taken (sec) | Filters used | Notes |
|---|---|---|---|---|---|---|

Keep this sheet for every participant.

---

## 🎯 Step 5. The Actual Task

You'll run the **same task** for all participants.

### Task Script (read verbatim)

> "You are planning a holiday from **London to Rome** between **10 and 17 December**.
> Your budget is **£250**, and you would prefer a flight with **no more than one stopover**.
> Please use Google Flights to find a flight that fits this budget and preference.
> Let me know once you believe you've found a suitable option."

---

## 🧍‍♂️ Step 6. During Each Session

### 1️⃣ Welcome & Consent

- Greet participant, explain what will happen.
- Give them the info sheet + consent form.
- Ask them to confirm understanding.
- Assign a participant ID (e.g., P01).

### 2️⃣ Pre-task Questions (for classification)

Ask:

- “Roughly how many times have you booked flights online?”
    - 0–2 → novice
    - 3–10 → intermediate
    - 10+ → expert

Record their category.

## 3️⃣ Set Up Recording

- Open OBS and start screen recording (capture screen + audio if allowed).
- Ensure Google Flights homepage is open.

## 4️⃣ Give Task Instructions

Read the task aloud (as above).
Remind them:

> “Please think aloud if you can — just tell me what you’re looking for or thinking about.”

## 5️⃣ Start Timing

- Start stopwatch (or note OBS timestamp) the moment they click **‘Search’** after entering flight details.

## 6️⃣ Observe Quietly

- Don’t interfere — only help if they get completely stuck.
- Note down:
    - Which filters they use (Stops, Airlines, Times, Price, Bags, etc.).
    - Any hesitation or confusion.

- Errors or backtracking.

### 7️⃣ Stop Timing

- When they say “I found one,” stop the stopwatch and recording.
- Record **end time**, calculate **time taken** in seconds.
- Count and note **number of filters used** (based on their actions on screen).

### 8️⃣ Short Post-task Questions

After the task, ask:

- “How easy did you find it to use the filters?” (1–5 Likert scale)
- “Do you usually use filters when booking flights?” (Yes/No)

Record these answers (they add qualitative context).

---

## 🕐 Step 7. Test Duration

Each participant session will take **25–30 minutes total**, broken down as:

- 5 min → Welcome, consent, pre-task questions
- 10–12 min → Task and observation
- 5 min → Post-task questionnaire
- 5–8 min → Buffer/setup for next participant

You’ll likely spread data collection across **4–6 days** to reach your 20+ participants.

---

## 📊 Step 8. Data Storage

After each session:

1. Save OBS recording (rename as P01_LondonRome.mp4).

Enter data in your spreadsheet:

ID | Experience | Time(sec) | FiltersUsed | Notes

2.
3. Save all in a secure folder (no real names).

4. Store consent forms separately (paper or scanned).

This becomes your **raw data** file for the report submission.

---

## 📈 Step 9. After All Sessions – Analysis

When all data is collected:

1. Group data by **experience level**.

2. Compute **average number of filters** used per group.

3. Run a **One-Way ANOVA** (you can use SPSS, Excel, or Python).

    - $H_0$: $\mu_1 = \mu_2 = \mu_3$ (no difference)

    - $H_1$: at least one group mean differs

4. If significant → do a **Tukey post-hoc** to see which groups differ.

---

## 📋 Step 10. Evidence to Include in the Report

In your submission folder, include:

- The observation sheet (scanned or typed)

- A few screenshots of participants using filters
- Sample recruitment email/message
- Screenshot of your spreadsheet
- Short recording clip (if allowed)
- Participant Information Sheet & Consent Form (Appendix I)

---

## ☑ Summary of Your Role (Saad)

| Phase | What you do |
|---|---|
| Design | Define ANOVA hypothesis & variables |
| Preparation | Create task instructions, forms, and observation sheet |
| Data Collection | Moderate in-person tests and record data |
| Data Management | Store files, anonymise IDs, update spreadsheet |
| Analysis | Run ANOVA, summarise results for the report |

---

Would you like me to make you a **ready-to-use Excel/CSV data sheet template** for this (with all the right columns and formulas for ANOVA later)?
That way you can start entering participant results directly.

# ANOVA

## 🧠 What ANOVA Means

**ANOVA** stands for **Analysis of Variance**.
It's a **statistical test** used to check **whether the means (averages)** of **three or more groups** are significantly different from each other.

Instead of doing multiple t-tests (which increases error risk), ANOVA compares all group means **in a single test** by looking at **the ratio of variance between groups to variance within groups**.

## ⚖️ The Basic Idea

ANOVA answers one main question:

> "Are the differences between the group means bigger than what we'd expect just by random chance?"

If the differences are large compared to the natural variation inside each group, then at least one group is **statistically different** from the others.

## 🧩 Example with Your Project (Google Flights)

Your research question is:

> "Do users with different experience levels (novice, intermediate, expert) differ in how many filters they use on Google Flights?"

Here's how it fits into ANOVA:

| Term | What it means in your study |
|---|---|
| **Independent Variable (Factor)** | User experience level (3 groups: novice, intermediate, expert) |
| **Dependent Variable** | Number of filters used during the flight search |

| | |
|---|---|
| **Goal** | See if the average number of filters used differs significantly across the 3 experience groups |

# 📊 How ANOVA Works (Step by Step)

Let's imagine you collected your data:

| Participant | Experience | Filters Used |
|---|---|---|
| P01 | Novice | 2 |
| P02 | Novice | 3 |
| P03 | Intermediate | 5 |
| P04 | Expert | 7 |
| ... | ... | ... |

Now ANOVA works like this:

## Step 1 — Calculate Group Means

- Mean of Novices = average filters used by novice users
- Mean of Intermediates = average filters used by intermediates
- Mean of Experts = average filters used by experts

## Step 2 — Calculate the "Overall Mean"

This is the mean of all data combined.

## Step 3 — Partition Variance

ANOVA breaks the total variability in your data into two parts:

1. **Between-group variance (SS_between)**
   → How far each group mean is from the overall mean
   → Measures the effect of your independent variable (experience level)

2. **Within-group variance (SS_within)**
   → How much participants within each group differ from each other
   → Measures random error or individual differences

### Step 4 — Compute the F-Ratio

[
F = \frac{\text{Variance Between Groups}}{\text{Variance Within Groups}}
]
If the groups are truly different, the numerator (between variance) will be large, so **F** will be large.

### Step 5 — Compare F to a Critical Value

Use an **F-distribution table** or software (like SPSS, Excel, or Python) to find a **p-value**.

- If **p < 0.05**, you reject the null hypothesis — meaning **at least one group is significantly different**.

- If **p ≥ 0.05**, you fail to reject it — meaning **no significant difference** between groups.

## 🔍 Step 6 — Post-Hoc Tests (If Significant)

If ANOVA shows a significant difference, you still don't know **which** groups differ (Novice vs Expert? Intermediate vs Expert?).
To find out, you run **post-hoc tests** such as **Tukey's HSD** or **Bonferroni**, which compare pairs of group means while controlling for error.

Example interpretation:

> ANOVA showed a significant difference in filter usage (p = 0.01).
> Post-hoc Tukey tests revealed that experts used significantly more filters than novices (p = 0.004), but intermediates did not differ from experts (p = 0.12).

## 🧮 What ANOVA Actually Compares (Conceptually)

| Type of Variance | Meaning |
|---|---|
| **Between-group variance** | Difference due to your independent variable (experience level) |
| **Within-group variance** | Random noise, personal differences, or measurement error |
| **F-ratio =** | Between variance ÷ Within variance |

If F ≈ 1 → groups are similar (no significant difference)
If F >> 1 → groups are likely different (significant result)

## 📜 ANOVA Assumptions (Important for Report)

To use ANOVA correctly, these **assumptions** must be met:

1. **Normality** — The dependent variable (e.g., filters used) is approximately normally distributed within each group.

2. **Homogeneity of variance** — Variance across groups should be roughly equal (test using Levene's test).

3. **Independence** — Participants in one group shouldn't influence another group's results.

If these are violated, you can use a **nonparametric alternative** such as the **Kruskal-Wallis test**.

## 💡 Interpreting in Your Case

If your one-way ANOVA gives:

- **F(2, 17) = 5.42, p = 0.012**,
  then you'd report:

> "A one-way ANOVA revealed a significant effect of user experience on the number of filters used, F(2,17) = 5.42, p = .012. Post-hoc analysis showed that expert users

applied significantly more filters than novice users, while differences between intermediate and expert users were not statistically significant."

## 🧰 Tools You Can Use

You can perform ANOVA in:

- **SPSS** → Analyze → Compare Means → One-Way ANOVA
- **Excel** → Data Analysis → ANOVA: Single Factor
- **Python (pandas/scipy)** → `f_oneway(group1, group2, group3)`
- **R** → `aov()` function

All will output:

- F-statistic
- Degrees of freedom (df_between, df_within)
- p-value

## 📜 Summary Table

| Concept | Your Study Example |
|---|---|
| **Test** | One-way ANOVA |
| **IV** | User experience level (novice/intermediate/expert) |
| **DV** | Number of filters used |
| **Purpose** | To determine if filter use differs by experience |
| **Hypotheses** | $H_0$: $\mu_1 = \mu_2 = \mu_3$; $H_1$: at least one mean differs |

**Statistic** F-ratio

**Follow-up** Tukey post-hoc test if $p < 0.05$